RAJALAKSHMI ENGINEERING COLLEGE

Soundaryan R 251801323

S2

Started on	Thursday, 24 September 2026, 10:24 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:45 AM
Time taken	21 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n, i, j;
    int arr[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("%d", arr[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 24 September 2026, 10:46 AM
State	Finished
Completed on	Friday, 9 October 2026, 3:39 PM
Time taken	15 days 4 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int n, i, j;
    int arr[100];
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("%d", arr[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 24 September 2026, 10:59 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:12 AM
Time taken	13 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
int T;
scanf("%d", &T);
while (T--) {
int n1, n2;
scanf("%d", &n1);
int a[n1];
for (int i = 0; i < n1; i++) {
scanf("%d", &a[i]);
}
scanf("%d", &n2);
int b[n2];
for (int i = 0; i < n2; i++) {
```

```
15  scanf("%d", &b[i]);
16  }
17  int i = 0, j = 0;
18  while (i < n1 && j < n2) {
19  if (a[i] == b[j]) {
20  printf("%d ", a[i]);
21  i++;
22  j++;
23  }
24  else if (a[i] < b[j]) {
25  i++;
26  }else {
27  j++;
28  }}
29  printf("\n");
30  }
31  return 0;
32  }
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 8 October 2026, 10:15 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:17 AM
Time taken	2 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
int T;
scanf("%d", &T);
while (T--) {
int n1, n2;
scanf("%d", &n1);
int a[n1];
for (int i = 0; i < n1; i++) {
scanf("%d", &a[i]);
}
scanf("%d", &n2);
int b[n2];
for (int i = 0; i < n2; i++) {
```

```
15 scanf("%d", &b[i]);
16 }
17 int i = 0, j = 0;
18 while (i < n1 && j < n2) {
19 if (a[i] == b[j]) {
20 printf("%d ", a[i]);
21 i++;
22 j++;
23 }
24 else if (a[i] < b[j]) {
25 i++;
26 }
27 else {
28 j++;
29 }}
30 printf("\n");
31 }
32 return 0;
33 }
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 8 October 2026, 10:47 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:56 AM
Time taken	8 mins 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
int n, k;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++) {
scanf("%d", &a[i]);
}
scanf("%d", &k);
int i = 0, j = 1;
while (i < n && j < n) {
if (i == j) {
j++;
}
else if (a[j] - a[i] == k) {
printf("1");
return 0;
}
else if (a[j] - a[i] < k) {
j++;
}
else {
i++;
}
}
printf("0");
return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 8 October 2026, 10:39 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:59 AM
Time taken	20 mins 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

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Output Format:

1 - If pair exists

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Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
int n, k;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++) {
scanf("%d", &a[i]);
}
scanf("%d", &k);
 int i = 0, j = 1;
while (i < n && j < n) {
if (i == j) {
j++;
}
else if (a[j] - a[i] == k) {
printf("1");
return 0;
}
else if (a[j] - a[i] < k) {
j++;
}else {
i++;
}}
printf("0");
return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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